

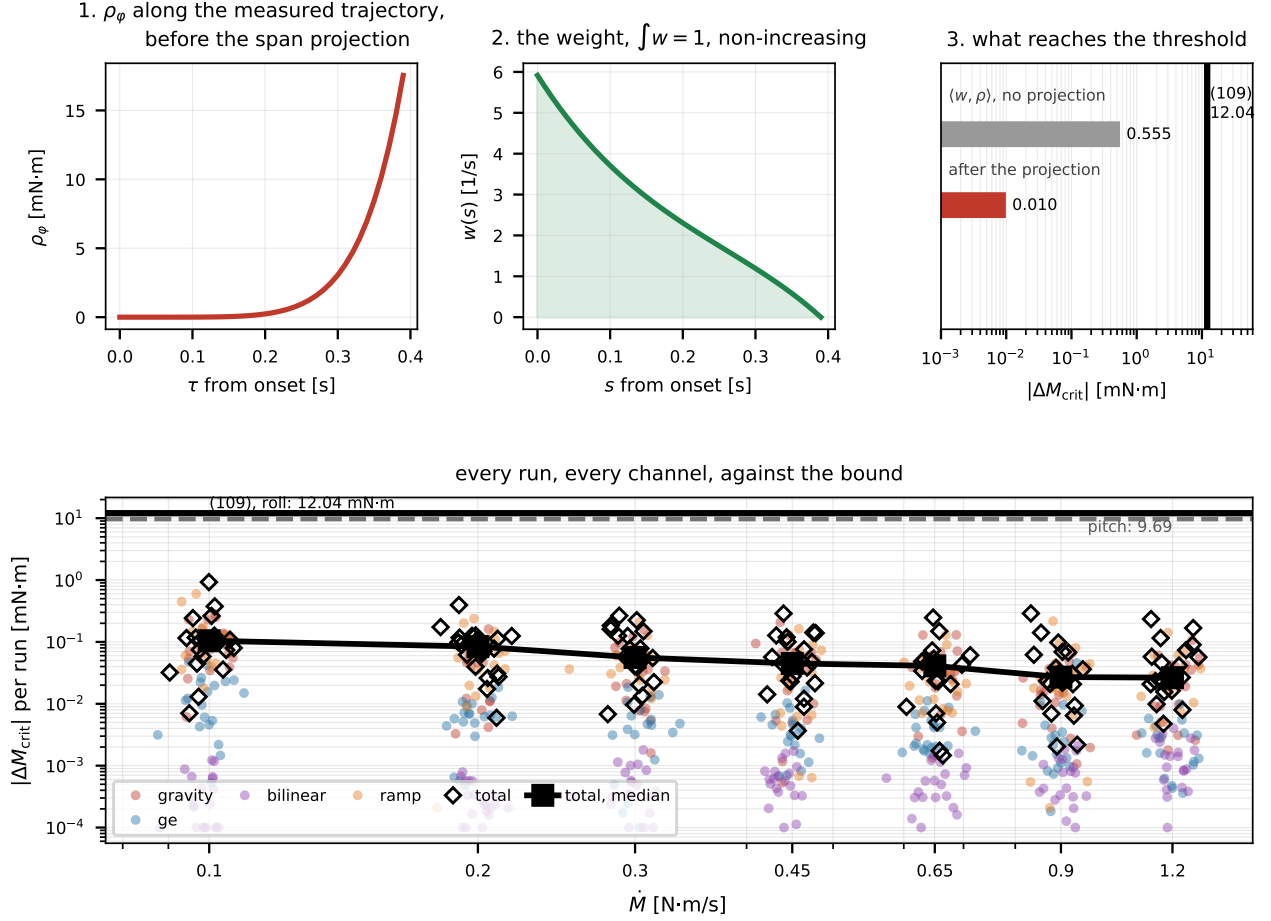


Figure 1: The a priori bound (109) checked forwards on the campaign. *Top, one run* (case 1, roll, $\dot{M} = 1.20$ N m/s): the gravity remainder ρ_φ evaluated along the measured trajectory; the normalised weight w of (107), which integrates to one and decreases, so that the forcing is largest exactly where the weight is least; and the resulting displacement, before and after the estimator’s own degrees of freedom are projected out, against the bound. *Bottom, all 140 runs*: the displacement per run and per channel. Every run lies inside, by a factor of 246 in the median and 13 at worst, and the gravity channel dominates as the bound predicts. The fit residual enters this chain nowhere; see the text.

0.1 The a priori bound against the campaign

Sec. VI-E bounds the displacement of the identified threshold by the two forcing terms the estimator cannot absorb, giving 0.400 mm of equivalent offset for roll and 0.322 for pitch over the admissible box. That bound is made before the campaign, from a scale, a CAD inertia and the excitation protocol; what remains is to ask what the runs actually do.

The question has to be put forwards. Each perturbation is evaluated along the *measured* trajectory over the fit window, projected out of the estimator’s own per-run degrees of freedom $\{1, \tau, \delta\varphi\}$, propagated through the Duhamel integral of the deviation dynamics, and correlated against the weight of (107). Fig. ?? shows that chain on one run and its result on all of them. The displacement is 0.049 mN m in the median, 0.161 at p90 and 0.927 at worst, against the 12.04 mN m the bound allows: every one of the 140 runs is inside, with two orders of margin typically and a factor of 13 in the worst case. The split across channels is the one the bound assumes, the gravity remainder supplying almost all of it and the ground-effect coupling two decades less.

Two features of the chain are worth reading off the top row. The forcing rises steeply towards the end of the window, as $\rho_\varphi \propto \varphi^2$ must, while the weight falls monotonically from the onset — the pairing that Chebyshev’s inequality turns into the factors $\frac{1}{7}$ and $\frac{1}{5}$, and the reason the bound is four times sharper than the envelope it replaces. And the projection is not a detail: on the run shown it takes the displacement from 0.555 to 0.010 mN m, the estimator absorbing what looks to it like a shifted onset.

Why the prediction residual is not this check. It is natural to compare the bound with the residual of Sec. VI-A instead, and the comparison is ill-posed rather than merely conservative. The bound is on the

deviation from the nominal response at the true onset; the residual is the deviation from a curve whose onset was chosen so as to minimise that residual. Whatever part of the deviation lies along $\partial\hat{\omega}/\partial t_c \propto \sinh C_2\tau$ is therefore removed by identity, so the residual is everything orthogonal to the estimator's span while the displacement bounded here is exactly the projection the residual discards. The two partition the deviation between them and neither carries information about the other. Pointwise the mismatch is conspicuous — the bound's envelope leaves the onset like τ^7 while the measurement starts at the gyro noise floor, so the residual sits outside it over 90% of the window at every ramp rate — and none of that bears on the bound. The residual is a prediction test on the model shape, reported as such in Sec. VI-A; the bound is checked here, forwards.